HCP-Bridge – Hoermann series 4 to Wi-Fi / Loxone

HCP bus and power

File: bus.kicad_sch

ESP32-S3

File: esp32.kicad_sch

One board per door. Plugs into the BUS port (RJ12) of the operator (ProMatic/SupraMatic series 4), emulates a UAP1-HCP module and exposes the door via HTTP/MQTT (firmware HCPBridgeMqtt by Gifford47, environment HCP_Giffordv2).

2 HCP bus and power
3 ESP32-S3

WORK IN PROGRESS – rev 0.1 has not been manufactured or tested yet.

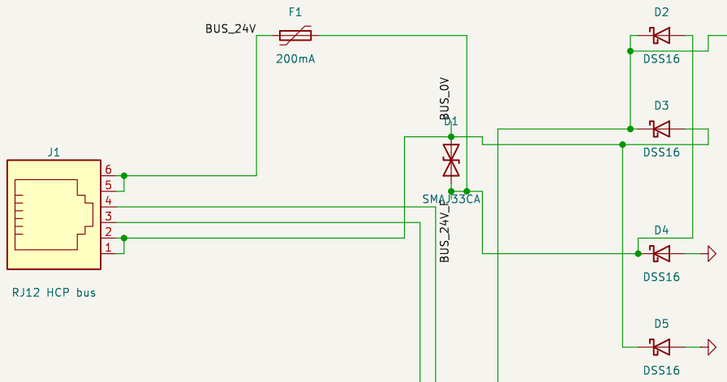
Schematic rev 0.1
Jonas Ernst

Sheet: /
File: HCPBridge.kicad_sch

Title: HCP-Bridge – overview

Size: A3	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 9.0.7		Id: 1/3

HCP bus: RJ12, fuse, TVS, bridge rectifier



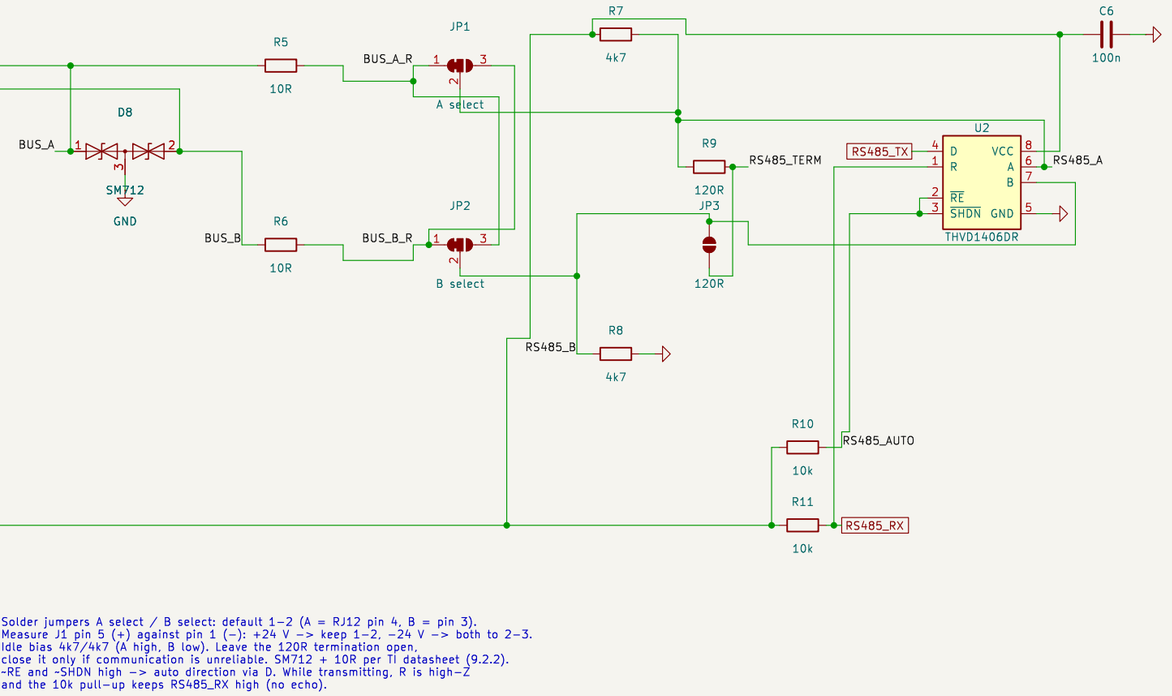
Bridge rectifier made of four Schottky diodes: the board works with straight and crossed RJ12 cables. Note: board GND sits one diode drop below bus GND (no problem for RS-485, common-mode range −7..12 V).
200 mA resettable fuse: the bridge draws about 20..40 mA from the 24 V bus.

USB power (flashing, bench testing)



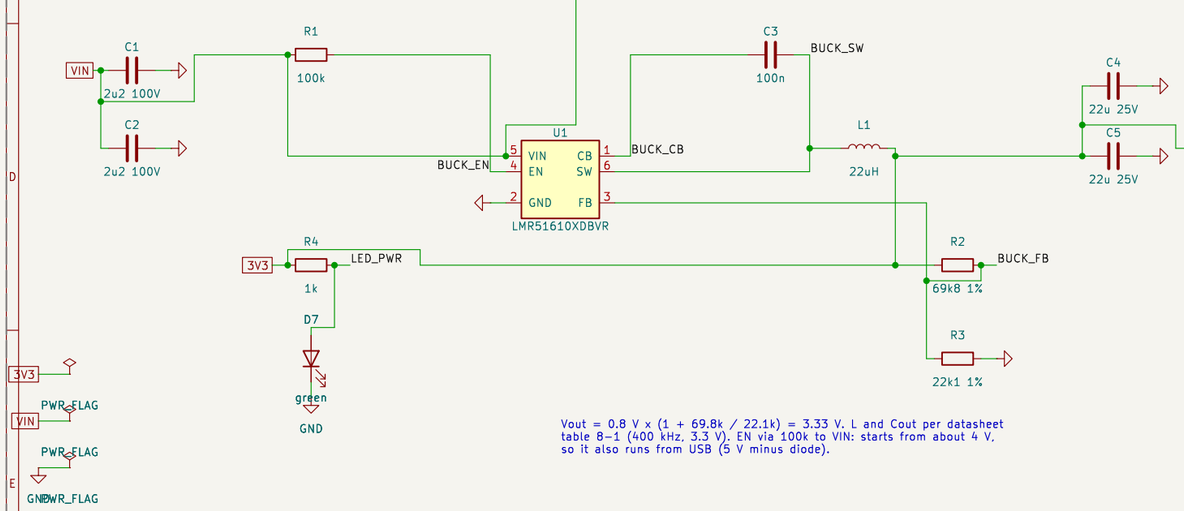
USB 5 V to VIN through a Schottky diode: the diodes prevent back-feeding between bus and USB.

RS-485 to the operator, THVD1406 (automatic direction control)



Solder jumpers A select / B select: default 1–2 (A = RJ12 pin 4, B = pin 3).
Measure J1 pin 5 (+) against pin 1 (−): +24 V −> keep 1–2, −24 V −> both to 2–3.
Idle bias 4k7/4k7 (A high, B low). Leave the 120R termination open,
close it only if communication is unreliable. SM712 + 10R per TI datasheet (9.2.2).
-RE and -SHDN high −> auto direction via D. While transmitting, R is high-Z
and the 10k pull-up keeps RS485_RX high (no echo).

3.3 V buck converter, LMR51610 (400 kHz, 1 A)



Vout = 0.8 V x (1 + 69.8k / 22.1k) = 3.33 V. L and Cout per datasheet table 8-1 (400 kHz, 3.3 V). EN via 100k to VIN: starts from about 4 V, so it also runs from USB (5 V minus diode).

RJ12, protection, rectifier, 3.3 V buck, RS-485

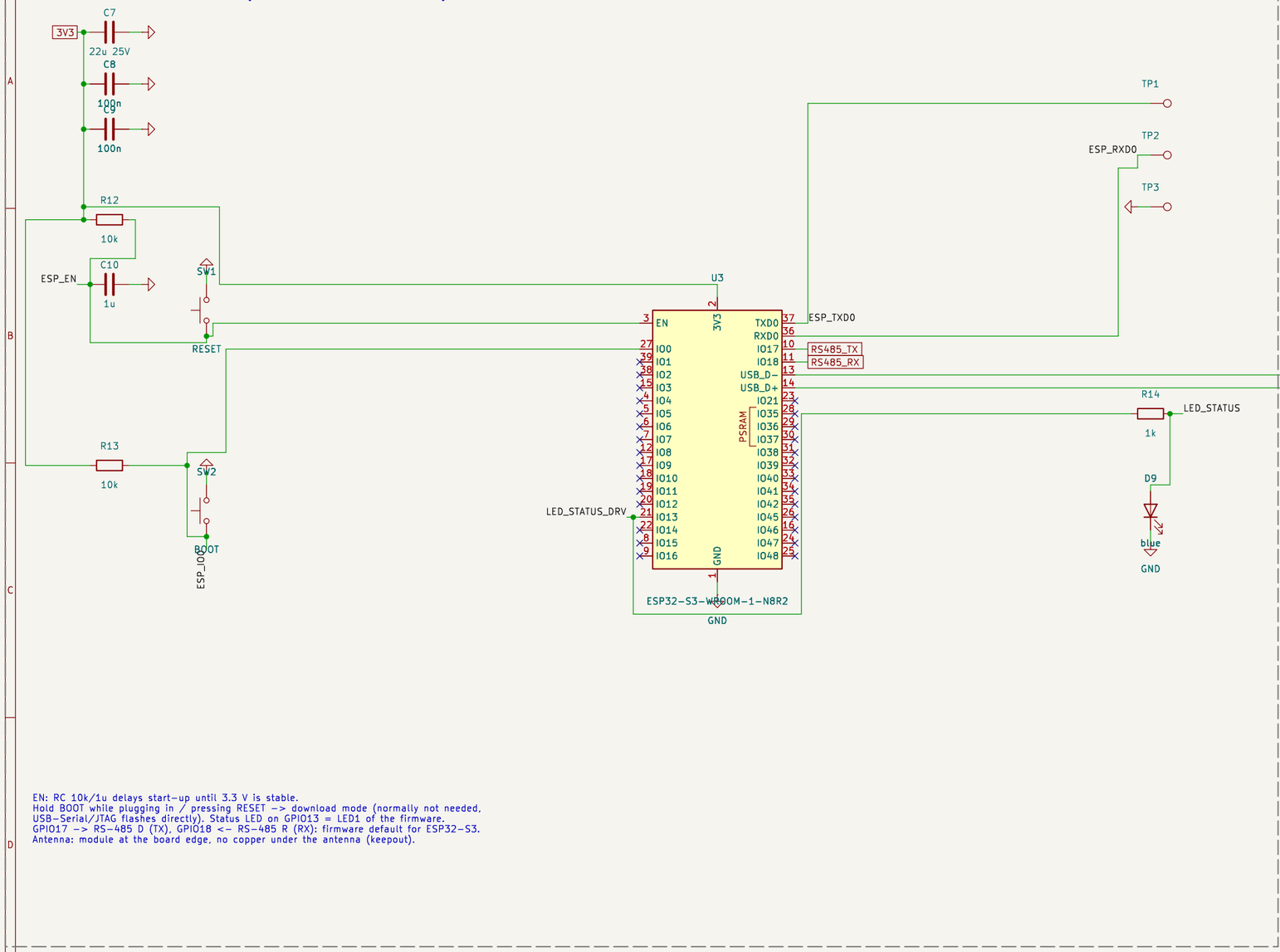
Jonas Ernst

Sheet: /HCP bus and power/
File: bus.kicad_sch

Title: HCP bus and power

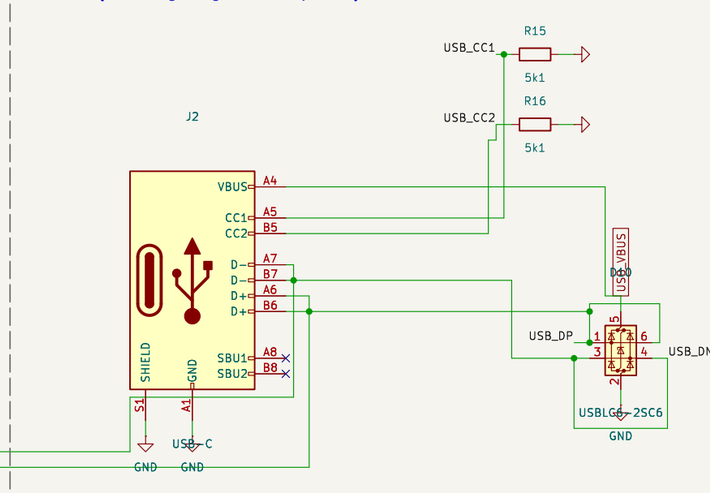
Size: A3	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 9.0.7		Id: 2/3

ESP32-S3-WROOM-1-N8R2 (Wi-Fi to Loxone / MQTT)



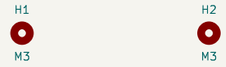
EN: RC 10k/1u delays start-up until 3.3 V is stable.
Hold BOOT while plugging in / pressing RESET -> download mode (normally not needed, USB-Serial/JTAG flashes directly). Status LED on GPIO13 = LED1 of the firmware.
GPIO17 -> RS-485 D (TX), GPIO18 <- RS-485 R (RX): firmware default for ESP32-S3.
Antenna: module at the board edge, no copper under the antenna (keepout).

USB-C (flashing, log, bench power)



CC 5k1 each to GND: sink, USB-C supplies provide 5 V.
Do not connect to a PC and the bus at the same time if PC and operator have different earthing (ground loop).

Mounting



ESP32-S3-WROOM-1, USB-C, buttons, LEDs

Jonas Ernst

Sheet: /ESP32-S3/
File: esp32.kicad_sch

Title: ESP32-S3

Size: A3	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 9.0.7		Id: 3/3